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Inventor(s)	Tadele; Wegene H. et al.

Sensing health parameters in wearable devices

Abstract

Embodiments of this disclosure are directed to a wearable device having a housing, a display, and a sensor system. The display is at least partially surrounded by the housing. The sensor system is housed at least partially in the housing. The sensor system includes a first sensor, a second sensor, and a controller. The first sensor is configured to contact a body part of a user and generate a first signal. The second sensor is configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal. The controller is configured to generate a resultant signal by removing noise from the first signal using the second signal, and determine a health parameter of the user from the resultant signal.

Inventors: **Tadele; Wegene H. (San Francisco, CA), Spencer; Maegan K. (Emerald Hills, CA), Brandt; Riley E. (Menlo Park, CA), Nangia; Siddharth (San Francisco, CA)**

Applicant: **Apple Inc.** (Cupertino, CA)

Family ID: **1000008777576**

Assignee: **Apple Inc. (Cupertino, CA)**

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Primary Examiner: Mahmood; Nadia A

Attorney, Agent or Firm: Brownstein Hyatt Farber Schreck, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a nonprovisional of, and claims the benefit under 35 U.S.C. § 119(e) of and priority to U.S. Provisional Patent Application No. 63/239,358, filed Aug. 31, 2021, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The described embodiments relate generally to wearable devices, systems, and methods thereof for determining a health parameter of a user. More particularly, the present embodiments relate to measuring health parameters with more accuracy and reliability using sensors in wearable devices.

BACKGROUND

(2) Sensors are included in many of today's electronic devices, including electronic devices such as phones, smartphones, computers (e.g., tablet computers or laptop computers), wearable electronic devices (e.g., electronic watches, smart watches, or health or fitness monitors), game controllers, navigation systems (e.g., vehicle navigation systems or robot navigation systems), earbuds, headphones, headsets, glasses, and so on. Sensors may variously sense the presence of objects, distances to objects, proximities of objects, movements of objects (e.g., whether objects are moving, or the speed, acceleration, or direction of movement of objects), compositions of objects, a biophysical condition or status of a user (e.g., a user's health or fitness), and so on.

(3) Given the wide range of sensor applications, any new development in the configuration or operation of a sensor can be useful. New developments that may be particularly useful are developments that reduce the cost, size, complexity, part count, or manufacture time of the sensor, or developments that improve the sensitivity, functionality, accuracy, or speed of sensor operation. It is increasingly desirable to have more accurate and reliable measurement sensors in wearable devices, and to have measurement sensors that are lower cost.

SUMMARY

(4) The term embodiment and like terms, e.g., implementation, configuration, aspect, example, and option, are intended to refer broadly to all of the subject matter of this disclosure and the claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the claims below. Embodiments of the present disclosure covered herein are defined by the claims below, not this summary. This summary is a high-level overview of various aspects of the disclosure and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter. This summary is also not intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this disclosure, any or all drawings, and each claim.

(5) Embodiments of this disclosure are directed to a wearable device having a housing, a display, and a sensor system. The display may be at least partially surrounded by the housing. The sensor system may be housed at least partially in the housing. The sensor system may include a first sensor, a second sensor, and a controller. The first sensor may be configured to contact a body part of a user and generate a first signal. The second sensor may be configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal. The controller may be configured to generate a resultant signal by removing noise from the first signal using the second signal, and determine a health parameter of the user from the resultant signal.

(6) Embodiments of this disclosure are also directed to a wearable device having a housing and a sensor system. The housing may include a main body and at least one stem extending from the main body. The sensor system may be housed at least partially in the housing. The sensor system

may include a first sensor, a second sensor, and a controller. The first sensor may be configured to sense an aspect of a body part of a user when the wearable device is worn and generate a first signal. The second sensor may be configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal. The controller may be configured to combine the first signal and the second signal, and determine a health parameter of the user using the combined first and second signals.

(7) Embodiments of this disclosure are also directed to a method of determining a health parameter of a user. The method may include receiving a first signal from a first sensor of an electronic device when the first sensor is in contact with a body part of the user. The method may further include receiving a second signal from a second sensor of the electronic device while the second sensor is spaced apart from the body part and sensing a mechanical wave of an ambient environment of the electronic device. The method may further include removing at least a portion of a noise signal from the first signal, using the second signal, to form a resultant signal. The method subsequently includes determining the health parameter of the user from the resultant signal.

(8) The above summary is not intended to represent each embodiment or every aspect of the present disclosure. Rather, the foregoing summary merely provides an example of some of the novel aspects and features set forth herein. The above features and advantages, and other features and advantages of the present disclosure, will be readily apparent from the following detailed description of representative embodiments and modes for carrying out the present invention, when taken in connection with the accompanying drawings and the appended claims. Additional aspects of the disclosure will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments, which is made with reference to the drawings, a brief description of which is provided below.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

(2) FIG. **1**A shows a perspective top view of an example wearable device having sensors for determining a health parameter of the user of the wearable device, according to aspects of the present disclosure;

(3) FIG. **1**B shows a perspective rear view of the example wearable device of FIG. **1**A, according to aspects of the present disclosure;

(4) FIG. **1**C shows a perspective side view of the example wearable device of FIG. **1**A, according to aspects of the present disclosure;

(5) FIG. **2** shows a perspective front view of another example wearable device having sensors for determining a health parameter of the user of the wearable device, according to aspects of the present disclosure;

(6) FIG. **3** shows a perspective front view of an example earbud having sensors for determining a health parameter of the user of the earbud, according to aspects of the present disclosure;

(7) FIG. **4** shows a perspective front view of an example pair of glasses having sensors for determining a health parameter of the user of the pair of glasses, according to aspects of the present disclosure;

(8) FIG. **5** shows a block diagram of a method of determining a health parameter of a user using the wearable devices of FIGS. **1-4**, according to aspects of the present disclosure; and

(9) FIG. **6** shows an example schematic representation of an electronic system in the wearable devices of FIGS. **1-4**, according to aspects of the present disclosure.

(10) The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures. Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.

(11) The present disclosure is susceptible to various modifications and alternative forms, and some representative embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

(12) Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

(13) Various embodiments are described with reference to the attached figures, where like reference numerals are used throughout the figures to designate similar or equivalent elements. The figures are not necessarily drawn to scale and are provided merely to illustrate aspects and features of the present disclosure. Numerous specific details, relationships, and methods are set forth to provide a full understanding of aspects and features of the present disclosure, although one having ordinary skill in the relevant art will recognize that these aspects and features can be practiced without one or more of the specific details, with other relationships, or with other methods. In some instances, well-known structures or operations are not shown in detail for illustrative purposes. The various embodiments disclosed herein are not necessarily limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are necessarily required to implement aspects and features of the present disclosure.

(14) For purposes of the present detailed description, unless specifically disclaimed, and where appropriate, the singular includes the plural and vice versa. The word “including” means “including without limitation.” Moreover, words of approximation, such as “about,” “almost,” “substantially,” “approximately,” and the like, can be used herein to mean “at,” “near,” “nearly at,” “within 3-5% of,” “within acceptable manufacturing tolerances of,” or any logical combination thereof. Similarly, terms “vertical” or “horizontal” are intended to additionally include “within 3-5% of” a vertical or horizontal orientation, respectively.

(15) Additionally, directional terminology, such as “top”, “bottom”, “upper”, “lower”, “front”, “back”, “over”, “under”, “above”, “below”, “left”, “right”, etc. is used with reference to the orientation of some of the components in some of the figures described below. Because components in various embodiments can be positioned in a number of different orientations, directional terminology is used for purposes of illustration only and is in no way limiting. The directional terminology is intended to be construed broadly, and therefore should not be interpreted to preclude components being oriented in different ways. These words are intended to relate to the equivalent direction as depicted in a reference illustration; as understood contextually from the object(s) or element(s) being referenced, such as from a commonly used position for the object(s) or element(s); or as otherwise described herein. Further, it is noted that the term “signal” means a waveform (e.g., electrical, optical, magnetic, mechanical or electromagnetic) capable of traveling

through a medium such as DC, AC, sinusoidal-wave, triangular-wave, square-wave, vibration, and the like.
(16) Also, as used herein, the phrase "at least one of" preceding a series of items, with the term "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase "at least one of" does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at a minimum one of any of the items, and/or at a minimum one of any combination of the items, and/or at a minimum one of each of the items. By way of example, the phrases "at least one of A, B, and C" or "at least one of A, B, or C" each refer to only A, only B, or only C; any combination of A, B, and C; and/or one or more of each of A, B, and C. Similarly, it may be appreciated that an order of elements presented for a conjunctive or disjunctive list provided herein should not be construed as limiting the disclosure to only that order provided.
(17) Embodiments of the disclosure are directed to wearable devices, systems, and methods of determining a health parameter of a user, using sensors in the wearable devices. In some embodiments, one or more optical sensors positioned on a watch, phone, cuff, earbud, or other device can be used to measure health metrics of a user. For example, heart rate and peripheral oxygen saturation (SpO2) can be measured using infrared (IR), red, and green light-emitting diodes (LEDs). In other embodiments, non-optical sensors, such as piezoelectric or acoustic sensors, can also be used to measure health metrics, and can enable continuous or on-demand health metric measuring or monitoring. Non-optical sensors are interesting because of their potential to reduce power consumption requirements beyond what is possible with optical sensors. This can be advantageous in that consumer products typically face challenges in terms of size, battery life, and so on. Any reduction in a device's power consumption requirements can be beneficial in terms of extending the device's use between battery charges, and/or enabling the device to operate with a battery having a reduced size or weight.
(18) As described herein, two or more piezoelectric polyvinylidene fluoride (PVDF) sensors and/or acoustic sensors may be integrated into a wearable device and used to obtain relevant signals for health sensing. The wearable device may be watches, phones, earbuds, glasses, tablet computers, sleep products (e.g., in-bed sensor mats), cuffs (e.g., blood pressure cuffs), chest straps, and the like. This arrangement works by measuring physiological health parameters and rejecting unrelated external noises that affect the signals associated with the physiological health parameters. This enables external noise interferences to be attenuated, thus improving signal to noise ratio (SNR) and enabling localized micro vibration sensing at low power consumption compared to optical sensors. Health metrics information can be extracted from sensed signals after subtracting out external aggressors (e.g., noise), and in some embodiments, applying filtering to an acquired signal, to determine a user's heart rate (HR), respiration rate, the presence of micro vibrations, and so on. In some embodiments, multiple types of sensors may be included in a device to improve SNR and/or reduce power consumption.
(19) Relevant sensors to improve signal quality include microphones, piezoelectric PVDF sensors, capacitive gap sensors, strain gauges, accelophones, and accelerometers. Such sensors may increase a sensing bandwidth, sensitivity, SNR, or dynamic range or linearity of the measured physiological health parameters. Such sensors may be used to acquire a user's HR, ballistocardiogram (BCG)/seismocardiogram (SCG), respiratory motion, lung sounds, and so on. A user's respiration rate may be measured using seismocardiography. Such sensors can be less susceptible to body perspiration, environmental acoustic noise, motion artifacts, and so on. In some embodiments, such sensors may be a companion to optical sensing and may be switched ON or OFF depending on an optical SNR.
(20) These and other wearable systems, devices, methods, and apparatus are described with reference to FIGS. **1**A-**6**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should

not be construed as limiting.
(21) FIGS. **1**A-**1**C show a perspective top view, a perspective rear view, and a perspective side view, respectively, of an example wearable device **100** having sensors for determining a health parameter of the user of the wearable device **100**. The wearable device **100** may be a smartwatch, a fitness monitor, a health diagnostic device, or alternatively, any type of wearable and portable device. The wearable device **100** includes a housing **102** including a main body **106**, and a band **104** (e.g., wrist band) attached to the housing **102**. The band **104** is configured to secure the wearable device **100** to a body part (e.g., an arm, wrist, leg, ankle, or waist) or skin of the user. In different embodiments, coupling mechanisms other than the band **104** may be used to secure the wearable device **100** to at least a portion of a body of the user or a stationary object.
(22) The housing **102** may include an input or selection device, such as a crown **118** or an input button **120**. The housing **102** at least partially surrounds a display **108**. In some embodiments, the housing **102** may include a sidewall **110**, which sidewall **110** may support a front cover **112** (FIG. **1**A) and/or a back cover **114** (FIG. **1**B). The front cover **112** may be positioned over the display **108**, and may provide a window through which the display **108** may be viewed. In some embodiments, the display **108** may be attached to (or abut against) the sidewall **110** and/or the front cover **112**. In alternative embodiments of the wearable device **100**, the display **108** may not be included and/or the housing **102** may have an alternative configuration.
(23) The display **108** may include one or more light-emitting elements including, for example, light-emitting elements that define a light-emitting diode (LED) display, organic LED (OLED) display, liquid crystal display (LCD), electroluminescent (EL) display, or other type of display. In some embodiments, the display **108** may include, or be associated with, one or more touch and/or force sensors that are configured to detect a touch and/or a force applied to a surface of the front cover **112**.
(24) In some embodiments, the sidewall **110** of the housing **102** may be formed using one or more metals (e.g., aluminum or stainless steel), polymers (e.g., plastics), ceramics, or composites (e.g., carbon fiber). The front cover **112** may be formed, for example, using one or more of glass, a crystal (e.g., sapphire), or a transparent polymer (e.g., plastic) that enables a user to view the display **108** through the front cover **112**. In some non-limiting embodiments, a portion of the front cover **112**, such as a perimeter portion of the front cover **112**, may be coated with an opaque ink to obscure components included within the housing **102**. In some embodiments, all of the exterior components of the housing **102** may be formed from a transparent material, and components within the device **100** may or may not be obscured by an opaque ink or opaque structure within the housing **102**.
(25) The back cover **114** may be formed using the same material(s) that are used to form the sidewall **110** or the front cover **112**. In some embodiments, the back cover **114** may be part of a monolithic element that also forms the sidewall **110**. In other embodiments, and as shown, the back cover **114** may be a multi-part back cover, such as a back cover **114** having a first back cover portion **114-1** attached to the sidewall **110** and a second back cover portion **114-2** attached to the first back cover portion **114-1**. The second back cover portion **114-2** may, in some embodiments, have a circular perimeter and an exterior surface **119** having an arcuate profile.
(26) The front cover **112**, back cover **114**, or first back cover portion **114-1** may be mounted to the sidewall **110** using fasteners, adhesives, seals, gaskets, or other components. The second back cover portion **114-2**, when present, may be mounted to the first back cover portion **114-1** using fasteners, adhesives, seals, gaskets, or other components.
(27) A display stack or device stack (hereafter referred to as a “stack”) including the display **108** may be attached to (or abutted against) an interior surface of the front cover **112** and extend into an interior volume of the wearable device **100**. In some embodiments, the stack may include a touch sensor (e.g., a grid of capacitive, resistive, strain-based, ultrasonic, or other type of touch sensing elements), or other layers of optical, mechanical, electrical, or other types of components. In some

embodiments, the touch sensor (or part of a touch sensor system) may be configured to detect a touch applied to an outer surface of the front cover **112**, such as a display surface of the wearable device **100**.

(28) In some embodiments, a force sensor (or part of a force sensor system) may be positioned within the interior volume below and/or to the side of the display **108** (and in some embodiments, within the device stack). The force sensor (or force sensor system) may be triggered in response to the touch sensor detecting one or more touches on the front cover **112**, or a location or locations of one or more touches on the front cover **112**, and may determine an amount of force associated with each touch, or an amount of force associated with the collection of touches as a whole. The force sensor (or force sensor system) may alternatively trigger operation of the touch sensor (or touch sensor system), or may be used independently of the touch sensor (or touch sensor system).

(29) The wearable device **100** includes a sensor system **116** that is at least partially housed in the housing **102**. The various sensors in the sensor system **116** are capable of acquiring physiological information from the wearer or user of the wearable device **100** (e.g., a heart rate, respiration rate, blood oxygenation level, a ballistocardiogram, a seismocardiogram, blood pressure level, a blood flow rate, a blood glucose level, a partial thromboplastin time, body temperature, etc.), or to determine a status of the wearable device **100** (e.g., whether the wearable device **100** is being worn or a tightness of the wearable device **100**). The different sensors in the sensor system **116** may include a piezoelectric PVDF sensor, a strain sensor, a piezoelectric sensor, a capacitive gap sensor, an acoustic sensor, a microphone, an accelophone, an image sensor, a heat sensor, a motion sensor, a proximity sensor, an accelerometer, a gyroscope, or a pressure transducer.

(30) In some non-limiting embodiments, one or more such sensors may include a number of electromagnetic radiation emitters, such as visible light and/or infrared (IR) emitters, and/or a number of electromagnetic radiation detectors, such as visible light and/or IR detectors, such as electromagnetic radiation detectors including any of the detector pixels described herein.

(31) In some embodiments, the wearable device **100** may have a port **122** (or set of ports) on a side of the housing **102** (or elsewhere), and an ambient pressure sensor, ambient temperature sensor, internal/external differential pressure sensor, gas sensor, particulate matter concentration sensor, or air quality sensor may be positioned in or near the port(s) **122**. Further, in some embodiments, the wearable device **100** may include an image sensor configured for biometric authentication of the user.

(32) The sensor system **116** in the wearable device **100** includes a first sensor **126** disposed on or attached to the back cover **114**, or alternatively, on an interior surface **104-1** of the band **104** so that the first sensor **126** is in contact of a body part or skin of the user. In non-limiting embodiments, the first sensor **126** is configured to detect a change in vibration, motion, or radiation associated with a physiological measurement of the user and generate a first signal based on the physiological measurement. In some embodiments, the first sensor **126** is a piezoelectric PVDF sensor or strain sensor. The first sensor **126** may alternatively be an optical sensor or other type of sensor. The first sensor **126** may emit or transmit signals through the housing **102** (or back cover **114**) and/or receive signals or sense conditions through the housing **102** (or back cover **114**).

(33) Additionally, a second sensor **128** is disposed on or attached to the front cover **112** of the housing **102**, or alternatively, on an exterior surface **104-2** of the band **104**. The second sensor **128** is configured to sense a mechanical wave in an ambient environment of the wearable device **100** and generate a second signal. In some embodiments, the second sensor **128** is embedded somewhere in the band **104** to be used as a reference to measure and reject external noise aggressors. In various embodiments, the second sensor **128** may be a piezoelectric PVDF sensor, a strain sensor, a capacitive gap sensor, an acoustic sensor, a microphone, an accelophone, and the like. In some embodiments, the second sensor **128** is embedded on the display **108** for self-examination and acquisition of other stethoscope sounds. In other embodiments, the second sensor **128** may be located at a remote location from the wearable device **100**, such as on a chest trap, on

blood pressure cuffs, or on an in-bed sensor mat.
(34) Additionally or alternatively, the sensor system **116** in the wearable device **100** may include multi-pixel PVDF, strain, or capacitive gap sensors **130**, **132**, **134**, and **136** embedded in or around the back cover **114** (e.g., around the second back cover portion **114-2**) to measure alternating current (AC) lung and heart signals. This arrangement can be used to measure partial thromboplastin time (PTT), respiration rate (RR), or heart rate (HR), by measuring and rejecting external noises through the multi-location sensor arrangement. This multi-pixel arrangement can also be used to detect whether the wearable device **100** is being worn and/or a tightness of the wearable device **100** while worn, or to detect a quality of contact between the skin of the user and the wearable device **100**, or to detect any rotation, skew, or an orientation of the wearable device **100**. For example, whether the wearable device **100** is being worn, or the quality of contact with a user's skin, can be determined based at least partly based on the strengths of signals generated by the sensors **130**, **132**, **134**, **136** and, in some cases, the amount of noise in the signals. As another example, whether the wearable device **100** is skewed with respect to a user's skin (e.g., not resting flat on the user's skin) can be determined at least partly based on comparisons of, and differences in, the signals generated by the sensors **130**, **132**, **134**, **136**. These detections of device condition(s) and orientation can be useful in conjunction with measurement of biometric signals (e.g., heart rate, respiration rate, BCG, etc.) and physiological signals because they can help identify issues with signal quality. For example, when measuring SpO2, detecting tightness of the wearable device **100**, or a quality of contact of the wearable device **100** with the user's skin is needed to accurately measure a perfusion index.
(35) Additionally or alternatively, the wearable device **100** may include a multi-pixel sensor arrangement including sensors **138**, **140** embedded into the back cover **114** and the band **104** (or, additionally or alternatively, two or more such sensors embedded in the band **104**).
(36) Additionally or alternatively, the sensor system **116** may include a temperature sensor (not shown) embedded on the back cover **114** for measuring a temperature of the wrist skin, and a temperature sensor (not shown) embedded on the front cover **112** for measuring an ambient temperature. The measured ambient temperature and the measured skin temperature may then be used to calculate a body temperature of the user (e.g., by adjusting a temperature acquired by the back cover temperature sensor in response to an ambient temperature measured by the front cover temperature sensor).
(37) The wearable device **100** may include a controller **124** (e.g., a processor and/or other components) configured to determine or extract, at least partly in response to signals received directly or indirectly from the sensors **126**, **128**, physiological health parameters of the user and/or a status of the wearable device **100**, as mentioned above. As a non-limiting example, the controller **124** may receive a first signal from the first sensor **126** comprising physiological information (e.g., heart rate, respiration rate, etc.) detected through contact of the user's skin. At the same time, the controller **124** may receive a second signal from the second sensor **128**, where the second signal includes external noise aggressors in the ambient environment of the wearable device **100**. The controller **124** uses the second signal to generate a resultant signal by removing noise captured by the first signal. In some embodiments, this is accomplished by simply subtracting the second signal from the first signal. In some embodiments, the controller **124** may further apply filters to the resultant signal to extract the health parameters or statuses.
(38) In some embodiments, the controller **124** may be configured to convey the determined or extracted health parameters or statuses via an output device of the wearable device **100**. For example, the controller **124** may cause the indication(s) to be displayed on the display **108**, indicated via audio or haptic outputs, transmitted via a wireless communications interface or other communications interface, and so on. The controller **124** may also or alternatively maintain or alter one or more settings, functions, or aspects of the wearable device **100**, including, in some embodiments, what is displayed on the display **108**.

(39) FIG. **2** shows a perspective front view of an example wearable device **200** having sensors for determining a health parameter of the user of the wearable device **200**. The wearable device **200** may be a smartphone, a mobile device, a tablet computer, a portable electronic device, a portable computer, a portable music player, a portable terminal, a vehicle navigation system, a robot navigation system, or other portable or mobile device. The wearable device **200** could also be a device that is semi-permanently located (or installed) at a single location (e.g., a door lock, thermostat, refrigerator, or other appliance).
(40) The wearable device **200** includes a housing **202** that at least partially surrounds a display **204**. The housing **202** may include or support a front cover **206** and/or a rear cover **208**. The front cover **206** may be positioned over the display **204**, and may provide a window through which the display **204** (including images displayed thereon) may be viewed by a user. In some embodiments, the display **204** may be attached to (or abut against) the housing **202** and/or the front cover **206**.
(41) The display **204** may include one or more light-emitting elements or pixels, and in some embodiments, may be an LED display, an OLED display, an LCD, an EL display, a laser projector, or another type of electronic display. In some embodiments, the display **204** may include, or be associated with, one or more touch and/or force sensors that are configured to detect a touch and/or a force applied to a surface of the front cover **206**.
(42) The various components of the housing **202** may be formed from the same or different materials. For example, a sidewall **218** of the housing **202** may be formed using one or more metals (e.g., stainless steel), polymers (e.g., plastics), ceramics, or composites (e.g., carbon fiber). In some embodiments, the sidewall **218** may be a multi-segment sidewall including a set of antennas. The antennas may form structural components of the sidewall **218**. The antennas may be structurally coupled (to one another or to other components) and electrically isolated (from each other or from other components) by one or more non-conductive segments of the sidewall **218**. The front cover **206** may be formed, for example, using one or more of glass, a crystal (e.g., sapphire), or a transparent polymer (e.g., plastic) that enables a user to view the display **204** through the front cover **206**. In some embodiments, a portion of the front cover **206** (e.g., a perimeter portion of the front cover **206**) may be coated with an opaque ink to obscure components included within the housing **202**. The rear cover **208** may be formed using the same material(s) that are used to form the sidewall **218** or the front cover **206**, or may be formed using a different material or materials. In some embodiments, the rear cover **208** may be part of a monolithic element that also forms the sidewall **218** (or in some embodiments, where the sidewall **218** is a multi-segment sidewall, those portions of the sidewall **218** that are non-conductive). In still other embodiments, all of the exterior components of the housing **202** may be formed from a transparent material, and components within the wearable device **200** may or may not be obscured by an opaque ink or opaque structure within the housing **202**.
(43) The front cover **206** may be mounted to the sidewall **218** to cover an opening defined by the sidewall **218** (i.e., an opening into an interior volume in which various electronic components of the wearable device **200**, including the display **204**, may be positioned). The front cover **206** may be mounted to the sidewall **218** using fasteners, adhesives, seals, gaskets, or other components.
(44) A display stack or device stack (hereafter referred to as a “stack”) including the display **204** (and in some embodiments, the front cover **206**) may be attached to (or abutted against) an interior surface of the front cover **206** and extend into the interior volume of the wearable device **200**. In some embodiments, the stack may also include a touch sensor (e.g., a grid of capacitive, resistive, strain-based, ultrasonic, or other type of touch sensing elements), or other layers of optical, mechanical, electrical, or other types of components. In some embodiments, the touch sensor (or part of a touch sensor system) may be configured to detect a touch applied to an outer surface of the front cover **206** (e.g., to a display surface of the wearable device **200**).
(45) In some embodiments, a force sensor (or part of a force sensor system) may be positioned within the interior volume below and/or to the side of the display **204** (and in some embodiments,

within the stack). The force sensor (or force sensor system) may be triggered in response to the touch sensor detecting one or more touches on the front cover **206** (or indicating a location or locations of one or more touches on the front cover **206**), and may determine an amount of force associated with each touch, or an amount of force associated with the collection of touches as a whole.
(46) The wearable device **200** includes a sensor system **216** that is at least partially housed within the housing **202**. The various sensors in the sensor system **216** are capable of acquiring physiological information from the wearer or user of the wearable device **200** (e.g., a heart rate, a respiration rate, a blood oxygenation level, a ballistocardiogram, a seismocardiogram, a blood pressure level, a blood flow rate, a blood glucose level, a partial thromboplastin time, etc.), or to determine a status of the wearable device **200** (e.g., whether the wearable device **200** is being worn or a tightness of the wearable device **200**). The different sensors in the sensor system **216** may include a piezoelectric PVDF sensor, a strain sensor, a piezoelectric sensor, a capacitive gap sensor, an acoustic sensor, a microphone, an accelophone, an image sensor, a heat sensor, a motion sensor, a proximity sensor, an accelerometer, a gyroscope, or a pressure transducer.
(47) In some embodiments, one or more such sensors may include a number of electromagnetic radiation emitters (e.g., visible light and/or IR emitters) and/or a number of electromagnetic radiation detectors (e.g., visible light and/or IR detectors, such as electromagnetic radiation detectors including any of the detector pixels described herein). In some embodiments, the wearable device **200** may have an ambient pressure sensor, an ambient temperature sensor, an internal/external differential pressure sensor, a gas sensor, a particulate matter concentration sensor, or an air quality sensor positioned at one or more locations on the housing **202**. Further, in some embodiments, the wearable device **200** may include an image sensor configured for biometric authentication of the user.
(48) In some embodiments, the wearable device **200** may include a first sensor **226** integrated into or with the display **204**, for purposes of monitoring a user's health. In some embodiments, the first sensor **226** may be capable of sensing physiological information from a body part or skin of a user of the wearable device **200**. In non-limiting embodiments, the first sensor **226** is configured to detect a change in vibration, motion, or radiation associated with a physiological measurement of the user and generate a first signal based on the physiological measurement. In some embodiments, the first sensor **226** may be a PVDF sensor or a strain sensor. The first sensor **226** may emit or transmit signals through the housing **202** and/or receive signals or sense conditions through the housing **202**.
(49) Additionally, a second sensor **228** is disposed on or attached to the rear cover **208** of the housing **202**. The second sensor **228** is configured to sense a mechanical wave in an ambient environment of the wearable device **200** and generate a second signal. In some embodiments, the second sensor **228** may be used as a reference to measure and reject external noise aggressors. In various embodiments, the second sensor **228** may be a PVDF sensor, a strain sensor, a piezoelectric sensor, a capacitive gap sensor, an acoustic sensor, a microphone, an accelophone, and the like. In some embodiments, the second sensor **228** is embedded on the display **108** for self-examination and acquisition of other stethoscope sounds. In other embodiments, the second sensor **228** may be located at a remote location from the wearable device **200**, such as on a chest trap, on blood pressure cuffs, or on an in-bed sensor mat.
(50) The wearable device **200** may include a controller **224** (e.g., a processor and/or other components) configured to determine or extract, at least partly in response to signals received directly or indirectly from the sensors **226**, **228**, physiological health parameters of the user, a status of the wearable device **200**, or parameters of an environment of the wearable device **200** (e.g., air quality), or a composition of a target or object, as mentioned above. As a non-limiting example, the controller **224** may receive a first signal from the first sensor **226** comprising physiological information (e.g., heart rate, respiration rate, etc.) detected through contact of the

user's skin. At the same time, the controller **224** may receive a second signal from the second sensor **228**, where the second signal includes external noise aggressors in the ambient environment of the wearable device **200**. The controller **224** uses the second signal to generate a resultant signal by removing noise captured by the first signal. In some embodiments, this is accomplished by simply subtracting the second signal from the first signal. In some embodiments, the controller **224** may further apply filters to the resultant signal to extract the health parameters or statuses.

(51) In some embodiments, the controller **224** may be configured to convey the determined or extracted health parameters or statuses via an output device of the wearable device **200**. For example, the controller **224** may cause the indication(s) to be displayed on the display **204**, indicated via audio or haptic outputs, transmitted via a wireless communications interface or other communications interface, and so on. The controller **224** may also alternatively maintain or alter one or more settings, functions, or aspects of the wearable device **200**, including, in some embodiments, what is displayed on the display **204**.

(52) The wearable device **200** may include various other components. For example, the front of the wearable device **200** may include one or more front-facing cameras **210** (including one or more image sensors), speakers **212**, microphones **214** or other audio components (e.g., audio, imaging, and/or sensing components) that are configured to transmit or receive signals to/from the wearable device **200**. In some embodiments, a front-facing camera **210**, alone or in combination with other sensors, may be configured to operate as a bio-authentication or facial recognition sensor. Additionally or alternatively, the sensor system **216** may be configured to operate as a front-facing camera **210**, a bio-authentication sensor, or a facial recognition sensor.

(53) The wearable device **200** may also include input buttons or other input devices positioned along the sidewall **218** and/or on a rear surface of the wearable device **200**. For example, a volume button or multipurpose button **220** may be positioned along the sidewall **218**, and in some embodiments, may extend through an aperture in the sidewall **218**. The sidewall **218** may include one or more ports **222** that allow air, but not liquids, to flow into and out of the wearable device **200**. In some embodiments, one or more sensors may be positioned in or near the port(s) **222**. For example, an ambient pressure sensor, ambient temperature sensor, internal/external differential pressure sensor, gas sensor, particulate matter concentration sensor, or air quality sensor may be positioned in or near a port **222**.

(54) FIG. **3** shows a perspective front view of another example wearable device **300** having sensors for determining a health parameter of the user of the wearable device **300**. The wearable device **300** may be an earbud, an earphone, and the like. In the embodiment shown in FIG. **3**, the wearable device **300** may include a housing **308** (i.e., an earbud housing) having a main body **310**, and a stem **304** extending from the main body **310**. In other embodiments (not shown), the wearable device **300** may have two parts, each of which is insertable in a user's ear and includes a housing with a main body and a stem. The housing **308** may hold a speaker **302** that can be inserted into a user's ear, an optional microphone **312**, and controller **306** that can be used to acquire audio from the microphone **312**, transmit audio to the speaker **302**, and communicate audio between the speaker **302**, the microphone **312**, and one or more remote devices. The controller **306** may communicate with a remote device wirelessly (e.g., via a wireless communications interface, using a Wi-Fi, BLUETOOTH®, or cellular radio communications protocol, for example) or via one or more wires (e.g., via a wired communications interface, such as a Universal Serial Bus (USB) communications interface). In addition to communicating audio, the controller **306** may transmit or receive instructions and so on.

(55) In some embodiments, the wearable device **300** may include one or more sensors **314** (e.g., PVDF sensor or strain sensor) integrated into the housing **308** for opportunistic/passive health monitoring. In non-limiting embodiments, the sensors **314** are configured to detect a change in vibration, motion, or radiation associated with a physiological measurement of the user and generate a first signal based on the physiological measurement. In some embodiments, a first one

of the sensors **314** may be disposed on the main body **310**, while a second one of the sensors **314** may be disposed on the stem **304**. In other embodiments, all of the sensors **314** may be either located on the stem **304** or on the main body **310**.

(56) The controller **306** may include a processor and/or other components that are configured to determine or extract, at least partly in response to signals received directly or indirectly from the speaker **302**, the microphone **312**, the one or more of the sensors **314**, information related to a proximity of a user, an input of a user, and so on. As a non-limiting example, the controller **306** may receive a first signal from one of the sensors **314** comprising physiological health parameters detected through contact of the user's skin, as described above. At the same time, the controller **306** may receive a second signal from the microphone **312**, where the second signal includes external noise aggressors in the ambient environment of the wearable device **300**. The controller **306** may use the second signal to generate a resultant signal by removing noise captured by the first signal. In some embodiments, this may be accomplished by simply subtracting the second signal from the first signal. In some embodiments, the controller **306** may further apply filters to the resultant signal to extract the health parameters. In some embodiments, the first signal may be received from a PVDF sensor, strain sensor, acoustic sensor, or temperature sensor **314** on the main body **310** and in contact with a user's skin, and the second signal may be received from a PVDF sensor, strain sensor, acoustic sensor, or temperature sensor **314** on the stem **304** and in contact with an ambient environment of a user's ear.

(57) In some embodiments, a user may wear a pair of the wearable devices **300**, in the form of separate earbuds, or different earphone housings of a headset, and so on. In these embodiments, environmental and/or physiological signals specific to each side of a user's head (e.g., each year) can be measured. In such embodiments, an accurate and precise clock/crystal embedded within each wearable device **300** or housing **308** can be used to synchronize and align time-stamped and/or time series measurements obtained from the sensors **314** of different wearable devices or housings. As an example, this can enable measurement of wind direction by detecting the relative magnitude of the received signals from sensors **314** positioned on different wearable devices **300** or housings **308** on a user's ears (e.g., PVDF sensors embedded on or inside the housing **308** on each of a user's ears or on each side of the user's head). For example, timings of changes in the signals produced by the sensors **314** can indicate air movement (e.g., pressure changes or acoustic propagation). As another example, this can enable measurement of temperature between the two ears by measuring the heat flux between the two ears using temperature sensors (e.g., sensors **314**) in contact with the user (e.g., flexible thermopile sensors embedded inside or on the surface of the housing **308** on each of a user's ears or on each side of the user's head), and extrapolation of the core body temperature of the user.

(58) In some embodiments, the controller **306** may be configured to convey the determined or extracted health parameters or statuses via an output device of the wearable device **300**. For example, the controller **306** may cause the indication(s) to be output via the speaker **302** or a haptic device, transmitted via a wireless communications interface or other communications interface, and so on. The controller **306** may also or alternatively maintain or alter one or more settings, functions, or aspects of the wearable device **300**, including, in some embodiments, what is output via the speaker **302**.

(59) FIG. **4** shows a perspective front view of yet another example wearable device **400** having sensors for determining a health parameter of the user of the wearable device **400**. The wearable device **400** may be a pair of glasses, smartglasses, and the like. By way of example, the wearable device **400** is shown to include a housing **402** and lenses **418**, **420**. The housing **402** includes a frame or main body **406** formed by a first lens rim **408**, a second lens rim **410**, and a bridge **404** connecting the first lens rim **408** to the second lens rim **410**. Both a first stem **412** connected to the first lens rim **408** and a second stem **414** connected to the second lens rim **410** extend from the main body **406**.

(60) Each of the first lens rim **408** and the second lens rim **410** may hold a respective lens, such as the first lens **418** or the second lens **420**. The lenses **418**, **420** may or may not magnify, focus, or otherwise alter light passing through the lenses **418**, **420**. For example, the lenses **418**, **420** may correct a user's vision, block bright or harmful light, or simply provide a physical barrier through which light can pass with no or minimal adjustment. In some embodiments, the first and second lenses **418**, **420** may be formed of glass or plastic.
(61) In some embodiments, a multi-pixel sensor system **416** may include a first sensor **422** and a second sensor **424** mounted to the wearable device **400**. The first sensor **422** and the second sensor **424** may be respectively mounted to the stems **412**, **414** or to other components of the wearable device **400**. In non-limiting embodiments, the first sensor **422** is configured to detect a change in vibration, motion, or radiation associated with a physiological measurement of the user and generate a first signal based on the physiological measurement. In non-limiting embodiments, the second sensor **424** is configured to sense a mechanical wave in an ambient environment of the wearable device **400** and generate a second signal that may be used as a reference to measure and reject external noise aggressors. The first sensor **422** and the second sensor **424** may, in some embodiments, be connected to a controller **426** that monitors a user's activity or health.
(62) The controller **426** may include a processor and/or other components that are configured to determine or extract, at least partly in response to signals received directly or indirectly from the first sensor **422** and the second sensor **424**, information related to a proximity of a user, an input of a user, and so on. As a non-limiting example, the controller **426** may receive a first signal from the first sensor **422** comprising physiological health parameters detected through contact of the user's skin, as described above. At the same time, the controller **426** may receive a second signal from the second sensor **424**, where the second signal includes external noise aggressors in the ambient environment of the wearable device **400**. The controller **426** uses the second signal to generate a resultant signal by removing noise captured by the first signal. In some embodiments, this is accomplished by simply subtracting the second signal from the first signal. In some embodiments, the controller **426** may further apply filters to the resultant signal to extract the health parameters.
(63) In some embodiments, the controller **426** may be configured to convey the determined or extracted health parameters or statuses via an output device of the wearable device **400**. For example, the controller **426** may cause the indication(s) to be output via a remote device or a haptic device, transmitted via a wireless communications interface or other communications interface, and so on.
(64) FIG. **5** shows a block diagram **500** of a method of determining a health parameter of a user using the wearable devices **100**, **200**, **300**, and **400** of FIGS. **1**A-**4**. The method starts at block **510**, where a first signal from a first sensor of an electronic device is received. The first sensor may be positioned in contact with a body part or a portion of a skin of the user. In some embodiments, the first sensor is configured to detect a change in vibration, motion, or radiation associated with a physiological measurement of the user and generate the first signal based on the physiological measurement. In some embodiments, the first sensor may be a polyvinylidene fluoride (PVDF) sensor, as described above.
(65) In the next step, in block **520**, a second signal from a second sensor of the electronic device is received. The second sensor is spaced apart from the body part of the user and configured to sense a mechanical wave of an ambient environment of the electronic device. In some embodiments, the second sensor may be a polyvinylidene fluoride (PVDF) sensor, as described above. In some embodiments, the second signal may be generated at the second sensor in response to receiving a mechanical wave through the ambient environment of the electronic device. This second signal may be used as a reference to measure and reject external noise aggressors.
(66) In block **530**, at least a portion of a noise signal from the first signal from the first sensor is removed, using the second signal from the second sensor, to form a resultant signal. In some embodiments, removing at least the portion of the noise signal from the first signal may include

subtracting the second signal from the first signal. In some embodiments, the controller **426** may further apply filters to the resultant signal to extract the health parameters.

(67) Finally, in block **540**, a health parameter of the user is determined from the resultant signal. In some embodiments, the health parameter of the user is indicative of a health condition of the user such as a heart rate, a respiration rate, a blood oxygenation level, a ballistocardiogram, a seismocardiogram, a blood pressure level, a blood flow rate, a blood glucose level, a partial thromboplastin time, and the like. In some embodiments, the health parameters may be subsequently outputted through a display, an audio output, a haptic output, a wirelessly transmitted message, and the like.

(68) FIG. **6** shows an example schematic representation of an electronic device **600** that is substantially similar to the wearable devices **100**, **200**, **300**, and **400** of FIGS. **1**A-**4**. The electronic device **600** may include an optional electronic display **602** (e.g., a light-emitting display), a processor **604**, a power source **606**, a memory **608** or storage device, a sensor system **610**, or an input/output (I/O) mechanism **612** (e.g., an input/output device, input/output port, or haptic input/output interface). The processor **604** may control some or all of the operations of the electronic device **600**. The processor **604** may communicate, either directly or indirectly, with some or all of the other components of the electronic device **600**. For example, a system bus or other communication mechanism **614** can provide communication between the electronic display **602**, the processor **604**, the power source **606**, the memory **608**, the sensor system **610**, and the I/O mechanism **612**.

(69) The processor **604** may be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions, whether such data or instructions is in the form of software or firmware or otherwise encoded. For example, the processor **604** may include a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), a controller, or a combination of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements. In some embodiments, the processor **604** may provide part or all of the controllers described with reference to FIGS. **1**A-**4**.

(70) It should be noted that the components of the electronic device **600** can be controlled by multiple processors. For example, select components of the electronic device **600** (e.g., the sensor system **610**) may be controlled by a first processor and other components of the electronic device **600** (e.g., the electronic display **602**) may be controlled by a second processor, where the first and second processors may or may not be in communication with each other.

(71) The power source **606** can be implemented with any device capable of providing energy to the electronic device **600**. For example, the power source **606** may include one or more batteries or rechargeable batteries. Additionally or alternatively, the power source **606** may include a power connector or power cord that connects the electronic device **600** to another power source, such as a wall outlet.

(72) The memory **608** may store electronic data that can be used by the electronic device **600**. For example, the memory **608** may store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences, timing signals, control signals, and data structures or databases. The memory **608** may include any type of memory. By way of example only, the memory **608** may include random access memory, read-only memory, Flash memory, removable memory, other types of storage elements, or combinations of such memory types.

(73) The electronic device **600** includes the sensor system **610**, including sensors positioned almost anywhere on the electronic device **600**. In some embodiments, the sensor system **610** may include one or more electromagnetic radiation emitters and/or detectors, positioned and/or configured as described with reference to any of FIGS. **1**A-**4**. The sensor system **610** may be configured to sense

one or more types of parameters, such as but not limited to, vibration; light; touch; force; heat; movement; relative motion; biometric data (e.g., physiological parameters) of a user; air quality; proximity; position; connectedness; matter type; and so on. By way of example, the sensor system **610** may include one or more of (or multiple of) a heat sensor, a position sensor, a proximity sensor, a light or optical sensor (e.g., an electromagnetic radiation emitter and/or detector), an accelerometer, a pressure transducer, a gyroscope, a magnetometer, a health monitoring sensor, an air quality sensor, and so on. Additionally, the sensor system **610** may utilize any suitable sensing technology, including, but not limited to, interferometric, magnetic, pressure, capacitive, ultrasonic, resistive, optical, acoustic, piezoelectric, PVDF, accelophone, or thermal technologies.
(74) The I/O mechanism **612** may transmit or receive data from a user or another electronic device. The I/O mechanism **612** may include the electronic display **602**, a touch sensing input surface, a crown, one or more input buttons (e.g., a graphical user interface "home" button), one or more cameras (including an under-display camera), one or more microphones or speakers, one or more ports such as a microphone port, and/or a keyboard. Additionally or alternatively, the I/O mechanism **612** may transmit electronic signals via a communications interface, such as a wireless, wired, and/or optical communications interface. Examples of wireless and wired communications interfaces include, but are not limited to, cellular and Wi-Fi communications interfaces.
(75) The foregoing description, for purposes of explanation, uses specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art, after reading this description, that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art, after reading this description, that many modifications and variations are possible in view of the above teachings.
(76) As described above, one aspect of the present technology may be the gathering and use of data available from various sources, including biometric data. The present disclosure contemplates that, in some instances, this gathered data may include personal information data that uniquely identifies or can be used to identify, locate, or contact a specific person. Such personal information data can include, for example, biometric data and data linked thereto (e.g., demographic data, location-based data, telephone numbers, email addresses, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information).
(77) The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to authenticate a user to access their device, or gather performance metrics for the user's interaction with an augmented or virtual world. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.
(78) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing

access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to some health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

(79) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of advertisement delivery services, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide data to targeted content delivery services. In yet another example, users can select to limit the length of time data is maintained or entirely prohibit the development of a baseline profile for the user. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

(80) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in some health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

(81) Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content can be selected and delivered to users by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the content delivery services, or publicly available information.

(82) Although the disclosed embodiments have been illustrated and described with respect to one or more implementations, equivalent alterations and modifications will occur or be known to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In addition, while a particular feature of the invention may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

(83) While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only, and not limitation. Numerous changes to the disclosed embodiments can be made in accordance with the disclosure herein,

without departing from the spirit or scope of the disclosure. Thus, the breadth and scope of the present disclosure should not be limited by any of the above described embodiments. Rather, the scope of the disclosure should be defined in accordance with the following claims and their equivalents.

Claims

1. A wearable device, comprising: a housing; a display at least partially surrounded by the housing; and a sensor system housed at least partially in the housing, the sensor system including, a first sensor configured to contact a body part of a user and generate a first signal; a second sensor configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal; and a controller configured to, generate a resultant signal by removing noise from the first signal using the second signal; and determine a health parameter of the user from the resultant signal, wherein: the second sensor comprises a piezoelectric sensor.
2. The wearable device of claim 1, wherein the housing defines a smartwatch.
3. The wearable device of claim 1, wherein the housing defines a smartphone.
4. The wearable device of claim 1, wherein the health parameter of the user includes at least one of: a heart rate; a respiration rate; a blood oxygenation level; a ballistocardiogram; a seismocardiogram; a blood pressure; a blood flow rate; a blood glucose level; or a partial thromboplastin time.
5. The wearable device of claim 1, wherein the sensor system includes a plurality of same types of sensors including the first sensor and the second sensor.
6. A wearable device, comprising: a housing, including, a main body; and at least one stem extending from the main body; and a sensor system housed at least partially in the housing, the sensor system including, a first sensor configured to sense an aspect of a body part of a user when the wearable device is worn, the first sensor generating a first signal; a second sensor configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal; and a controller configured to, combine the first signal and the second signal; and determine a health parameter of the user using the combined first and second signals, wherein: the second sensor comprises a polyvinylidene fluoride (PVDF) sensor.
7. The wearable device of claim 6, wherein: the wearable device is an earbud; the at least one stem is one stem; the first sensor is disposed on the main body; and the second sensor is disposed on the one stem.
8. The wearable device of claim 6, wherein: the main body includes a glasses frame; and the at least one stem includes a pair of stems extending from the glasses frame.
9. The wearable device of claim 6, wherein at least one of the first sensor or the second sensor is disposed on the at least one stem.
10. A method of determining a health parameter of a user, the method comprising: receiving a first signal from a first sensor of an electronic device, the first sensor in contact with a body part of the user; receiving a second signal from a second sensor of the electronic device, the second sensor spaced apart from the body part, and the second sensor sensing a mechanical wave of an ambient environment of the electronic device; removing at least a portion of a noise signal from the first signal, using the second signal to form a resultant signal; and determining the health parameter of the user from the resultant signal, wherein: the first sensor comprises a first polyvinylidene fluoride (PVDF) sensor; the second sensor comprises a second PVDF sensor; and the removing the at least the portion of the noise signal from the first signal comprises subtracting the second signal from the first signal.
11. The method of claim 10, wherein: the first PVDF sensor is attached to a back cover of a housing a wearable device; and the second PVDF sensor is attached to a front cover of the housing of the wearable device.

12. The method of claim 10, further comprising: generating the second signal at the second sensor in response to receiving the mechanical wave through the ambient environment of the electronic device.

13. The method of claim 10, wherein: the first PVDF sensor is disposed on an interior surface of a band of a wearable device; and the second PVDF sensor is disposed on an exterior surface of the band of the wearable device.

14. A wearable device, comprising: a housing; a display at least partially surrounded by the housing; and a sensor system housed at least partially in the housing, the sensor system including, a first sensor configured to contact a body part of a user and generate a first signal; a second sensor configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal; and a controller configured to, generate a resultant signal by removing noise from the first signal using the second signal; and determine a health parameter of the user from the resultant signal, wherein: the second sensor comprises a capacitive gap sensor.

15. A wearable device, comprising: a housing; a display at least partially surrounded by the housing; and a sensor system housed at least partially in the housing, the sensor system including, a first sensor configured to contact a body part of a user and generate a first signal; a second sensor configured to sense a mechanical wave in an ambient environment of the wearable device and generate a second signal; and a controller configured to, generate a resultant signal by removing noise from the first signal using the second signal; and determine a health parameter of the user from the resultant signal, wherein: the second sensor comprises an acoustic sensor.